How temperature regimes near the equinox synchronize spring biological events

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Abstract

Many biological processes, including plant leafout and flowering, occur once a thermal-sum reaches a threshold. In this way, cumulative temperatures are thought to coordinate the timing of biological events within and between species. But a growing number of reports have shown that as climates warm, both the advancement of spring has slowed (declining sensitivity) and the variance in the timing of spring events has increased (declining synchrony)—raising questions about the resilience of temperature-based coordination to anthropogenic climate change. To account for these reports, researchers have greatly complicated the thermal-sum model, introducing additional factors and mechanisms. We consider whether such complexity is necessary. Using results from the theory of stopped random walks, we demonstrate that these reports are exactly as predicted by the basic thermal-sum model. The theory suggests that the relationship between temperatures and both the timing and synchrony of biological events is nonlinear. In particular, it predicts that as temperatures increase and springtime events shift from the equinox toward the solstice, springtime events become less coordinated and more variable. We verify these findings empirically, and we conclude that efforts to complicate the basic thermal-sum model should be consistent with the relevant theory.

Introduction

Many biological events are triggered not by the temperature on a single day, but by the temperature accumulated over many days. A canonical example is the law of the flowering plants, which states that a bud blooms in the spring once cumulative temperatures exceed a plant-specific threshold. This and similar mechanisms synchronize a wide variety of seasonal events in ecosystems—for instance, ensuring that pollinators emerge when flowers are in bloom. They also support an array of forecasting tools in which scientists track cumulative temperature or 'growing degree days' to predict harvest dates, manage pest emergence, and assess how shifting temperature patterns reshape ecological communities with climate change.

Shifts to earlier biological events—i.e., earlier phenology—has been the clearest fingerprint of anthropogenic climate change across both natural and managed systems. But these shifts have proven difficult to fully predict or explain. In recent decades, the rate of advancement has decelerated in that biological events advance less with each degree of warming (9). At the same time, the variance in the timing of phenological events has in some cases declined and in others increased, without a clear pattern (10; 11). Higher variance has also manifested as previously unknown or rare phenological events becoming more common (12). While research has provided myriad explanations for these events—often with contrasting predictions and logic (9; 13; 14)—no work has examined the extent to which these observations are consistent with the basic thermal-sum model.

Scientists have used a basic thermal-sum model for over two centuries, but have rarely if ever integrated the relevant mathematical tools developed to characterize its behavior—specifically results from renewal theory designed for analyzing stopped random walks. Here we draw on these results to predict how events triggered by a cumulative temperature sum will shift with warming. Our model explains both the deceleration observed in the advancement of springtime events as well as the shifts in the variance. In particular, we show the importance of the temperature regime under which the biological events take place. For example, we show that the variance in the timing of biological events can decrease or increase with warming depending on whether temperatures during the period of accumulation are stationary (as in the winter and summer near the solstice) or increasing (as in the spring near the equinox). We conclude that climate change can (i) move spring events earlier at an increasing or decelerating rate, and (ii) increase the variance of their timings, producing less synchronized events as they arrive earlier, a phenomenon we refer to as a 'scattered spring.'

Results & Discussion

Many springtime events occur when cumulative temperatures first reach or 'hit' a plant-specific temperature threshold. Such hitting-time models fall within the domain of renewal theory, which we use here to approximate the distribution of event times (hitting times) and thus predict the behavior of temperature-triggered biological events.

Consider one such event, the bloom date of a bud. We model effective temperatures (X_i)—the temperatures experienced by the bud at a given location—in two temperature regimes, visualized in Figure 1. The first is the stationary temperature regime in which the average daily temperature ($\alpha > 0$) is constant. The stationary regime is commonly seen in the winter months following the solstice, and we refer to the temperatures in this regime as 'winter temperatures.' The second regime is the increasing temperature regime in which the average daily temperature increases at rate ($\beta > 0$). The increasing temperature regime is commonly seen in the spring months closer to the equinox, and we refer to the increase as 'spring warming.' In both regimes, the effective temperature is the daily average temperature plus noise (ϵ_i). That is, $X_i = \alpha + \epsilon_i$ in the stationary regime, and $X_i = \alpha + \beta\, i + \epsilon_i$ in the increasing regime.

Noise in this formulation could represent many biological aspects depending on the scale and our developing understanding of what triggers spring plant events. At the bud-level, it represents idiosyncratic variation at that bud, which could be driven potentially by microclimate variation (15), bud color variation (16), or internal differences that vary when the bud starts accumulating temperatures (CITESmolecularchillpapers). We assume the ϵ_i are independent with mean 0 and variance σ^2.

The bloom date $\nu(\tau)$ is the day (n) that the cumulative temperatures ($Z_n = \sum_{i=1}^{n} X_i$) first reach or 'hit' a plant-specific temperature-sum threshold $\tau > 0$ (i.e. $\nu(\tau) = \min\{m : Z_m > \tau\}$). This formulation is common in growing degree day models. Indeed, Z_n is the 'forcing' part of most process-based models of plant phenology (CITES).

Standard results for stopped random walks yield asymptotic normal approximations for the bloom date ($\nu(\tau)$) when the thermal-sum threshold (τ) is sufficiently large that accumulation takes place over many days. The mean and variance of the approximation depend crucially on the underlying temperature regime (i.e., values of α and β) during which the plant accumulates temperature. In the stationary regime, average temperatures are relatively flat during accumulation ($\beta \approx 0$), and cumulative temperature grows at an approximately linear rate. It follows that expected bloom date advances with climate change in step with increasing winter temperatures ($\mathbb{E}[\nu(\tau)] \approx \frac{\tau}{\alpha}$) as is often described in the literature (**?**). In the spring warming regime, average temperatures increase during accumulation ($\beta > 0$), and cumulative temperatures grow quadratically. In this regime, bloom date advances as a function of the rate of spring warming ($\mathbb{E}[\nu(\tau)] \approx \sqrt{\frac{2\tau}{\beta}}$).

Within either regime, the relationship between bloom dates and temperature is non-linear as observed in recent papers (9; 13). Increasing temperatures advance bloom dates at a diminishing rate as warming (measured by the winter temperatures, α, or spring warming, β) increases relative to the size of the threshold (for the stationary temperature regime: $\frac{\partial}{\partial \alpha}\,\mathbb{E}[\nu(\tau)] \approx -\frac{\tau}{\alpha^2}$; for the increasing temeperature regime: $\frac{\partial}{\partial \beta}\,\mathbb{E}[\nu(\tau)] \approx -\frac{1}{2}\sqrt{\frac{2\tau}{\beta^3}}$). But when climate change shifts the period when plants accumulate temperatures from the increasing temperature regime that holds near the spring equinox to the stationary temperature regime that holds near the winter solstice, advances may actually accelerate, particularly if increases in winter temperatures (α) outpace increases in spring warming (β).

Importantly, these two temperature regimes make very different predictions regarding the synchrony of events—that is the variance in event timings within the same year and location. Both predict decreased variance with warming (for the winter temperatures regime: $\mathrm{Var}(\nu(\tau)) \approx \frac{\sigma^2\tau}{\alpha^3}$; for the spring warming regime: $\mathrm{Var}(\nu(\tau)) \approx \frac{\sigma^2}{\beta^{3/2}\sqrt{2\tau}}$), but the influence of noise is strikingly different as reflected by whether the threshold (τ) is in the numerator or the denominator. In the increasing temperature regime—common near the equinoxes—the threshold (τ) is in the denominator, and thus the variance is small when the the threshold (τ) is large, reflecting the fact that noise is quickly overwhelmed by the rapid quadratic increase in cumulative temperatures that occurs during spring warming. This effectively acts to synchronize plants (and buds within plants) with the same threshold—making small differences in microclimate or other idiosyncratic effects negligible. In contrast, under the winter temperatures regime the threshold (τ) is in the numerator, and thus the variance is large when the threshold (τ) is large. The interpretation is that noise can remain

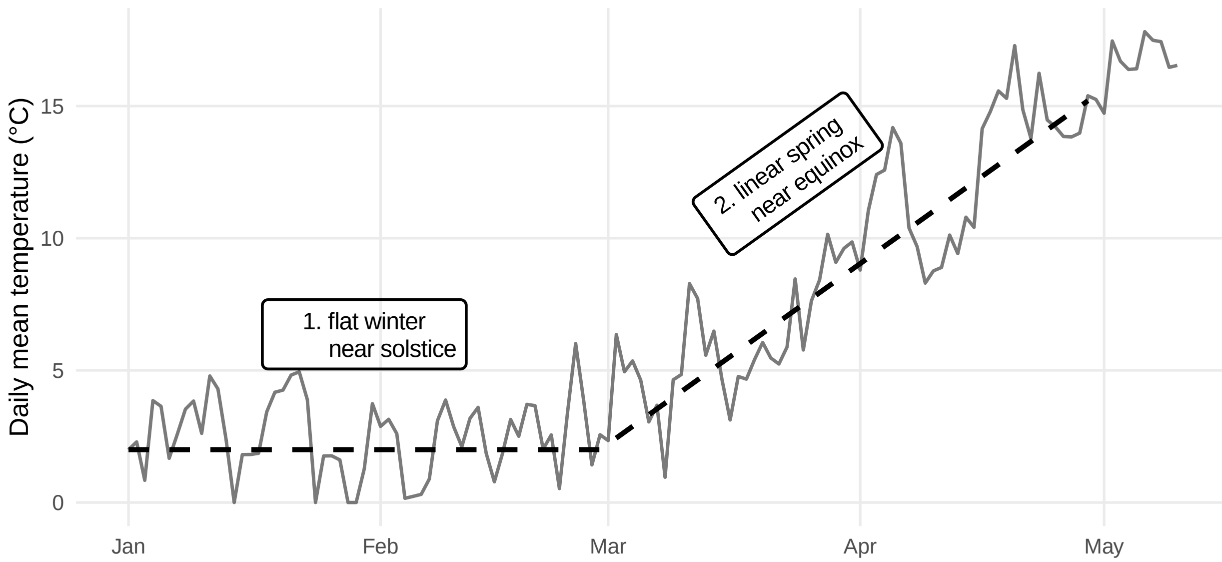

Figure 1: We model the daily temperature (solid line) as trend (dashed line) plus noise. The timing of springtime events in temperate climates depends on whether temperature is accumulated during one of two regimes, 1. the stationary temperature regime (relatively flat near the winter solstice), which we refer to as 'winter temperatures,' or 2. the increasing temperature regime (relatively linear near the spring equinox), which we refer to as 'spring warming.'

relatively influential so that small idiosyncratic differences—such as due to microclimate—translate into larger differences in threshold-crossing times. This prediction fits well with experiments conducted under constant temperature regimes, which find increasing variance with decreasing baseline temperatures (**?**). In such experiments, buds often have extremely variable budburst timings, as our model predicts (see Appendix).

The fact that the variance is predicted to be small in the spring warming case and larger in the winter temperatures case suggests the observed variance will increase when anthropogenic warming shifts accumulation from the spring warming regime to the winter temperatures regime. Indeed, as warming advances events earlier in the calendar, thresholds are reached in portions of the season with flatter mean temperatures where timing can be more sensitive to local variation. In such cases, warming produces a regime shift that simultaneously (i) advances springtime events potentially at a faster rate and (ii) reduces synchrony across nearby locations, producing a more variable and less coordinated onset of spring even as it arrives sooner, a phenomenon we refer to as a 'scattered spring.'

Whether warming leads to a more synchronized or more scattered spring depends on the local joint evolution of winter temperatures together with spring warming (α, β) and on the threshold t relevant for a given species and life-cycle event. In some locations, climate change may manifest primarily as higher winter temperatures (an increase in α) with relatively little change in the rate at which temperatures rise during spring (a modest change in spring warming, β). In others, winter temperatures may change less while spring transitions steepen (a larger increase in spring warming, β). The stopped-random-walk framework we present here offers a simple way to translate this spatially varying climate change into spatially varying predictions—with predictions about both the advancement of spring and its variability. To demonstrate this, we examine flowering dates data from the historical lilac–honeysuckle phenology network compiled by Rosemartin et al. (2015), collected across the continental United States from 1956–2014. We limit our analysis to the purple common lilac (*Syringa vulgaris*) and we update the dataset using USA NPN records from the same sites and species until 2025.

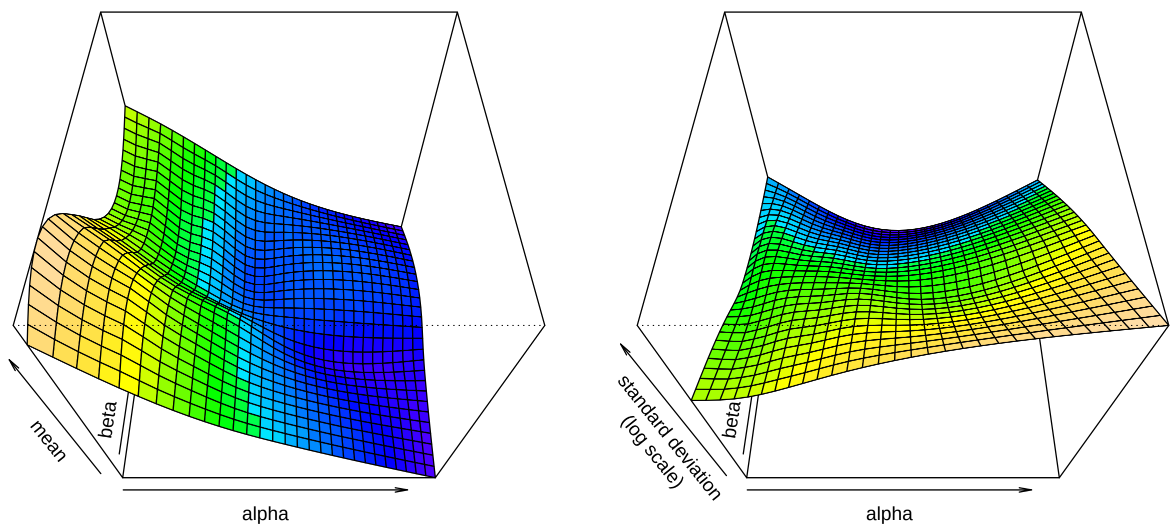

Figure 2: Mean and log standard deviation of lilac bloom dates (n = 10,613 from Rosemartin et al. 2015, updated to 2025 using USA NPN data) by baseline temperatures (α) and within-season temperature increases (β) estimated by a generalized additive model. The left figure shows the mean generally decreases as α and β increase. The standard deviation (right figure) can increase or decrease, again depending on the relative values of α and β.

We find that increasing winter temperatures and spring warming shift flowering times as predicted by our model in both shifts in mean bloom dates and their variances (Figure **??**, estimated nonparametrically as conditional means and standard deviations from a general additive model, GAM, see Hastie and Tibshirani (1986) and Wood (2017)). Looking only at the effect of spring warming (that is, picking any one spot along the winter temperatures, α, axis) shows that years with higher spring warming (higher β) are associated with earlier mean bloom dates, which generally advance at a decreasing rate. The same is true looking at only the effect of winter temperatures (that is, picking any one spot along the spring warming, β, axis). Consistent with the theory, the standard deviations (and thus the variances) generally declined with higher spring warming (i.e., as β increases holding α fixed, see right panel of Figure **??**).

However, in cases where winter temperatures increase, but spring warming does not, more buds may bloom under winter temperatures and thus the variation in bloom dates increases. This is the "scattered spring" predicted from our model. Warmer winters advance bloom dates in to an earlier portion of the seasonal cycle in which forcing is effectively flatter and cumulative temperatures grow more slowly. We find the same results

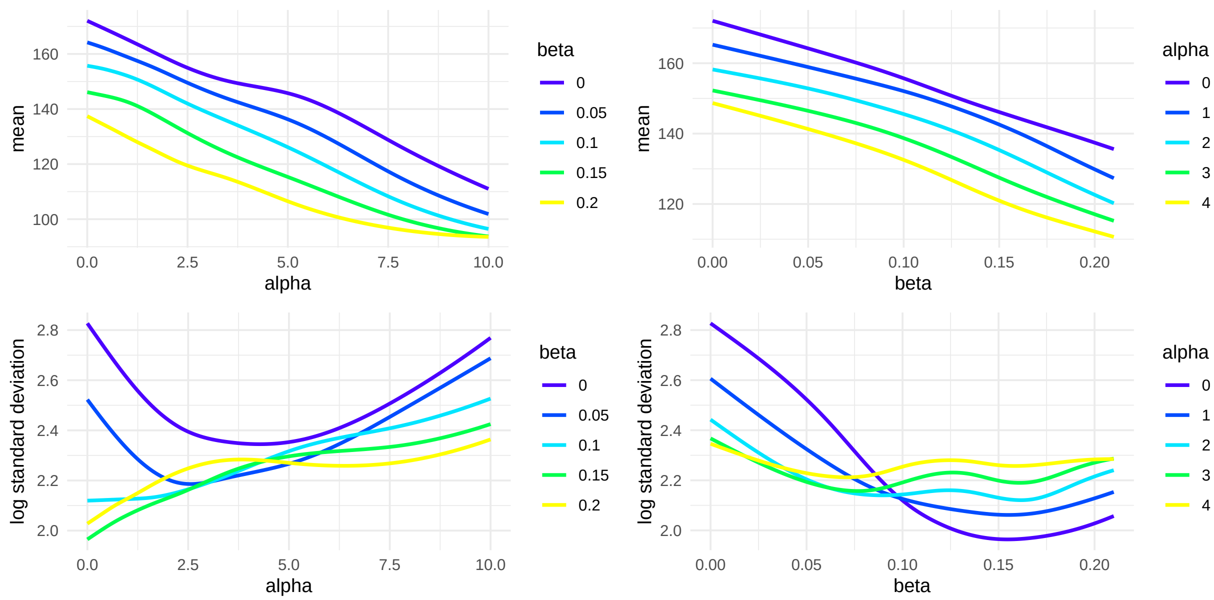

Figure 3: Mean (top) and log standard deviation (bottom) of lilac bloom dates (n = 10,613 from Rosemartin et al. 2015, updated to 2025 using USA NPN data) by baseline temperatures (α) and within-season temperature increases (β) estimated by a generalized additive model. The top two figures show the mean generally decreases as α increases holding β constant (left) and β increases holding α constant (right). The bottom two figures show the standard deviation can increase or decrease depending on whether α increases with β held constant (left) or β increases with α held constant (right)..

through an alternative approach where we bin sites and years within a grid of winter temperatures (α) and spring warming (β, see Appendix, Alternative approach to GAM').

Our results show that the basic thermal-sum model can explain the current suite of shifts in plant event timings, including how advances have slowed as warming continues and shifts of increasing and decreasing variances (10; 9; 11). This full suite of predictions comes from the most basic model for biological events that are triggered only by a thermal-sum—the same as most growing degree day models or the spring forcing component of many process-based plant models. While our model does not preclude alternative explanations, it is the most parsimonious. More complex models require additional justification. For example, winter chilling requirements or effects of photoperiod—which are often invoked to explain the trends in means and variance we reproduce with only a thermal-sum model—may be more useful, and testable, if they could make more unique predictions than they currently do.

The stopping-time framework shows current observations of shifting spring events follow from the thermal-sum model that has been used to predict bloom dates for over two centuries. While we show how it predicts the suite of observed shifts in the mean and variance of biological events, it could be further tested in experiments that vary stationary ('winter') temperatures and non-stationary ('spring) temperatures, and through continuing observational studies. In particular, the model predicts a general desynchronization of springtime events as their timing moves into the stationary regime closer to the solstice. If this occurs then increased warming would make predictions of biological event timings—from crops to forests, and from plants to insects—more variable, with potential cascading consequences for industry and natural ecosystems.

Methods

Our observational data analysis considers all sites in the historical lilac–honeysuckle phenology network that recorded the phenophase full bloom and were within 10 miles of a GHCND site with complete temperature records. We limit our analysis to (*Syringa vulgaris*), and include all observations made from 1955 to 2025 as recorded by the USA National Phenology Network, resulting in $10,613$ observations. For each observation, we estimate α using the average daily temperature between January and February. We estimate β using the slope of a linear regression model fit by regressing daily temperature on day index over March and April. Temperatures below 0 are set to 0 (a common base temperature in growing degree day models). The parameters α and β are shown for nine random sites in Figures 4 and 5 in Appendix A.2, which have the same interpretation as Figure 1. Site 14726 in New Mexico (Year 1969) is a good example of a bloom in the spring warming regime since relatively little temperature was accumulated during January and February. The average bloom date was May 8th with a standard deviation of 17 days. Site 14276 in New York (Year 1982) is a good example of a bloom between the winter temperatures and spring warming regims, which had an average bloom date of March 18 with a standard deviation of 27.

We fit a two-parameter Gaussian location–scale model using the `gaulss` function from `mgcv` where

$$\mu(\alpha, \beta) = \mathbb{E}[\nu(\tau) \mid \alpha, \beta] \quad \text{and} \quad \sigma(\alpha, \beta) = \mathrm{SD}(\nu(\tau) \mid \alpha, \beta).$$

Both μ and σ are modeled with tensor-product splines with basis dimension k = 10. The fitted functions are shown in Figures 2 and 3.

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Summary of analytical solution

To fix ideas, consider the day a bud blooms at a given location. We model the effective temperature experienced by that bud on day i by a linear trend plus noise,

$$X_i = \alpha + \beta\, i + \epsilon_i, \tag{1}$$

where $\alpha > 0$ is a baseline level, $\beta \geq 0$ is a within-season warming rate, and $\{\epsilon_i\}_{i\geq 1}$ are independent and identically distributed with $\mathbb{E}[\epsilon_i] = 0$ and $\mathrm{Var}(\epsilon_i) = \sigma^2$, representing idiosyncratic variation.

Let

$$Z_n \;=\; \sum_{i=1}^{n} X_i \qquad \text{and} \qquad \nu(t) \;=\; \min\{m : Z_m > t\} \tag{2}$$

denote cumulative temperature by day n and the observed bloom date, respectively, where $t > 0$ is a plant-specific tempearture-sum threshold. Thus $\nu(t)$ is the hitting time at which cumulative temperatures first exceed the threshold.

Standard results for stopped random walks yield asymptotic normal approximations for $\nu(t)$, with behavior governed by the parameters α and β. We consider two cases.

Regime 1: constant drift ($\beta = 0$) When average temperatures are positive and approximately constant,

$$\nu(t) \;\dot\sim\; \mathrm{Normal}\!\left(\frac{t}{\alpha},\; \frac{\sigma^2\, t}{\alpha^3}\right), \tag{3}$$

see Gut (2011) and Gut (2009, chap. 3).

Regime 2: increasing drift ($\beta > 0$) When average temperatures increase, the hitting time satisfies

$$\nu(t) \;\dot\sim\; \mathrm{Normal}\!\left(\sqrt{\frac{2t}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2},\; \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right), \tag{4}$$

with the derivation sketched in Appendix A.1 and simulations confirming the derivations in Appendix A.3. Note that when t is sufficiently large, the mean of $\nu(t)$ is approximately $\sqrt{2t\,/\,\beta}$.

The functional central limit theorem and linearization argument underlying these results hold somewhat generally, and thus similar limits can be derived when the $\{\epsilon_i\}_{i\geq 1}$ are weakly correlated or the temperature trajectory is not linear but sufficiently smooth.

From equations (3)–(4), we make two observations. Finding 1: there is a different nonlinear relationship between warming and the mean timing of springtime events, and thus warming can advance springtime events at an increasing rate. Finding 2: the two regimes are not equally sensitive to the idiosyncratic variation, and thus warming can increase the variation of springtime events, thereby scattering spring. We now discuss both findings in greater detail.

2.1 Finding 1: warming can advance springtime events at an increasing rate

In both regimes, the expected bloom date is a nonlinear function of the warming parameters.

Regime 1: constant drift ($\beta = 0$) From (3),

$$\mathbb{E}[\nu(t)] \approx \frac{t}{\alpha}, \qquad \mathrm{Var}(\nu(t)) \approx \frac{\sigma^2\, t}{\alpha^3}. \tag{5}$$

Increasing baseline temperatures (larger α) advances the expected bloom date, but the marginal advance diminishes since

$$\frac{\partial}{\partial \alpha}\, \mathbb{E}[\nu(t)] \approx -\frac{t}{\alpha^2}, \tag{6}$$

where the magnitude decreases as α grows.

Regime 2: increasing drift ($\beta > 0$) From (4), when t is large,

$$\mathbb{E}[\nu(t)] \approx \sqrt{\frac{2t}{\beta}}, \qquad \mathrm{Var}(\nu(t)) \approx \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}. \tag{7}$$

Increasing the within-regime warming rate (larger β) also advances the expected bloom date with diminishing returns since

$$\frac{\partial}{\partial \beta}\, \mathbb{E}[\nu(t)] \approx -\frac{1}{2}\sqrt{\frac{2t}{\beta^3}}. \tag{8}$$

However, if warming climates shift the bloom date from regime 2 to regime 1, then the rate of advancement can increase, particularly if increases in α outpace increases in β.

2.2 Finding 2: warming can scatter springtime events

Equations (3)–(4) also clarify how warming affects dispersion. Holding the regime fixed, warming reduces variance. That is, $\mathrm{Var}(\nu(t))$ decreases with α in (3) and with β in (4). However, just as in Section 2.1, warming can change which regime is relevant at the realized bloom date. When an event is advanced sufficiently early, idiosyncratic variation plays a larger role in determining the hitting time.

One way to quantify sensitivity to idiosyncratic variation is through the (squared) coefficient of variation,

$$\mathrm{CV}^2 \;=\; \frac{\mathrm{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2}.$$

In regime 1, $\mathbb{E}[\nu(t)] \approx t/\alpha$ and $\mathrm{Var}(\nu(t)) \approx \sigma^2 t/\alpha^3$, so

$$\mathrm{CV}_1^2 \;=\; \frac{\mathrm{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2} \;\approx\; \frac{\sigma^2 t/\alpha^3}{(t/\alpha)^2} \;=\; \frac{\sigma^2}{\alpha\, t}.$$

In regime 2, $\mathbb{E}[\nu(t)] \approx \sqrt{2t/\beta}$ and $\mathrm{Var}(\nu(t)) \approx \sigma^2/(\beta^{3/2}\sqrt{2t})$, hence

$$\mathrm{CV}_2^2 \;=\; \frac{\mathrm{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2} \;\approx\; \frac{\sigma^2/(\beta^{3/2}\sqrt{2t})}{(2t/\beta)} \;=\; \frac{\sigma^2}{\sqrt{\beta}\,(2t)^{3/2}}.$$

Both expressions increase when deterministic accumulation is weaker relative to σ (e.g., smaller α and/or smaller effective β). Consequently, warming tends to reduce variability within a given regime by strengthening deterministic forcing.

To compare regimes, note that

$$\frac{\mathrm{CV}_1^2}{\mathrm{CV}_2^2} \;\approx\; \frac{\sigma^2/(\alpha t)}{\sigma^2/(\sqrt{\beta}\,(2t)^{3/2})} \;=\; \frac{\sqrt{\beta}\,2^{3/2}}{\alpha}\sqrt{t}.$$

Thus, since α and β are nearly always greater than 0, the regime 1 becomes arbitrarily more sensitive to microclimate fluctuations than regime 1 for large t. This produces a more variable, less synchronized and thus "scattered" spring when warming shifts threshold crossings from regime 2 to regime 1.

Experimental Data

We first consider a controlled experiment on walnut trees (*Juglans*~sp.) conducted by Charrier et al. (2011). The data examines 30 trees comprising 6 genotypes at 2 locations. In November of the study year, stems were sampled from each tree and cut into 7 centimeter segments containing a single bud. All segments were chilled at 4°C and then transferred to constant 'forcing' environments with temperatures set at one of $5, 10, 15, 20$, or 25°C. For each temperature, the outcome is the response time, the time (in days) until budburst determined from the date at which 50% of buds reached stage 15 of the BBCH scale.

This experiment isolates the winter regime because temperature is held constant over time. Recall that in such cases, cumulative forcing grows approximately linearly,

$$Z_n \approx \alpha n + \sum_{i=1}^{n} \epsilon_i,$$

so that the hitting time $\nu(t)$ satisfies the approximation

$$\nu(t) \;\dot{\sim}\; \mathrm{Normal}\left(\frac{t}{\alpha},\, \frac{\sigma^2 t}{\alpha^3}\right),$$

implying that increasing α should (i) reduce the mean time to budburst and (ii) reduce dispersion. In addition, the squared coefficient of variation,

$$\mathrm{CV}^2 = \frac{\mathrm{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2},$$

provides a unit-free measure of sensitivity to stochastic fluctuations, which should decrease as α increases.

Table 1: Summary data from the Walnut forcing experiment (Charrier et al. 2011).

α	n	mean	sd	cv^2
5	30	178	27.28	0.0236
10	30	88	12.44	0.0199
15	30	39	11.33	0.0831
20	30	26	7.67	0.0851
25	30	20	4.45	0.0480

Table 1 reports the mean and standard deviation of the response times across the 30 stems at each forcing temperature, along with the sensitivity measure CV^2. The pattern is consistent with the regime 1 model. Higher forcing temperatures advance budburst at a decreasing rate (the mean falls with α), and the dispersion of budburst timing is smaller at warmer forcing temperatures. Thus, in an experimental setting, the stopped-random-walk approximation captures the basic comparative statics of constant-temperature forcing on both the level and variability of budburst times. (The sensitivty does decline from $\alpha = 20$ to $\alpha = 25$. The reason for low sensitivity for $\alpha = 5$ and 10 is unclear and may reflect censoring.)

Table 2: Mean bloom date (day-of-year) from lilac-honeysuckle data (Rosemartin et al. 2015) by average Jan-Feb temperature α (rows) and average Mar-Apr temperature increase β (columns).

$\alpha \setminus \beta$	$[-0.28, 0.07]$	(0.07, 0.12]	(0.12, 0.16]	(0.16, 0.68]
(0, 0.6]	163.05	154.24	148.96	142.56
(0.6, 1.6]	153.21	147.97	142.60	136.78
(1.6, 4]	138.86	133.56	130.06	126.34
(4, 16.4]	105.13	106.46	105.26	102.67

Table 3: Standard deviation of bloom date (day-of-year) from lilac-honeysuckle data (Rosemartin et al. 2015) by winter level α (rows) and spring warming rate β (columns).

$\alpha \setminus \beta$	$[-0.28, 0.07]$	(0.07, 0.12]	(0.12, 0.16]	(0.16, 0.68]
(0, 0.6]	16.08	12.36	10.44	9.61
(0.6, 1.6]	15.24	12.87	13.20	12.67
(1.6, 4]	17.97	16.52	16.31	14.56
(4, 16.4]	22.78	19.84	17.23	14.76

Alternative approach to GAM

In addition to the GAM we present in the main text, we also used a binning approach to estimate shifts in the mean and standard deviation of the historical lilac–honeysuckle phenology network compiled by Rosemartin et al. (2015). Both approaches address the same questions; we use both to make sure our results are consistent across approaches with slightly different underlying assumptions and biases. For the binning approach, we first bin sites and years within a grid of baseline temperatures (α, rows) and springtime warming temperatures (β, columns) and calculate the mean and standard deviation of the observed bloom date, as the number of days since January 1. within each bin (Tables 2 and 3 show binning approach).

First, looking at cases where the baseline temperature regime is relatively constant (that is, holding α fixed), shows that years with steeper springtime warming regimes (higher β) are associated with earlier mean bloom dates, which generally advance at a decreasing rate. The same is true in cases where the springtime warming regime is relatively constant (that is, holding β fixed), and the baseline temperature regime α increases. When baseline temperatures (α) are high, spring warming temperatures (β) appear to have little influence. This is predicted by the model since, under such conditions, plants accumulate sufficient temperature under the stationary regime, in which springtime warming temperatures (β) play no role, and thus bloom before reaching the springtime warming regime.

A. Appendix

A.1 Sketch of Theoretical Results

We follow the notation in Gut (2009, chap. 6). Let $X_i = \alpha + \beta i + \epsilon_i$ denote the temperature on day i, where $\alpha > 0$, $\beta \geq 0$, and the $\{\epsilon_i\}$ are independent and identically distributed with $\mathbb{E}[\epsilon_i] = 0$, and $\mathrm{Var}(\epsilon_i) = \sigma^2$. Define the deterministic cumulative sum

$$\xi_n = \sum_{i=1}^{n} \mathbb{E}[X_i] = \alpha n + \frac{\beta}{2} n(n+1) = \frac{\beta}{2} n^2 + \beta\gamma n \quad \text{where} \quad \gamma = \frac{\alpha}{\beta} + \frac{1}{2}$$

and $\xi_n = \alpha n$ when $\beta = 0$. Denote the sum of the stochastic component $S_n = \sum_{i=1}^{n} \epsilon_i$. Our object of interest is the first-passage time of $Z_n = S_n + \xi_n$,

$$\nu(t) = \min\{n \geq 1 : \; Z_n > t\}.$$

In the language of Lai and Siegmund (1977), $\{Z_n\}$ is a perturbed random walk: the random walk $\{S_n\}$ is perturbed by a deterministic term $\{\xi_n\}$. This setup yields two asymptotic regimes, corresponding to the empirical distinction between regime 1 ($\beta \approx 0$) and regime 2 ($\beta > 0$), which in turn underlies our qualitative predictions about nonlinear changes in themean and variance.

In regime 1, $\beta = 0$, and we apply the usual asymptotic argument for the hiting time of a random walk with constant positive drift as $t \to \infty$,

$$\nu(t) \mathrel{\dot\sim} \mathrm{Normal}\left(\frac{t}{\alpha}, \; \frac{\sigma^2 \, t}{\alpha^3}\right),$$

see Gut (2011) and Gut (2009, chap. 3).

In regime 2, $\beta > 0$, and the deterministic component $\xi_n = \frac{\beta}{2} n^2 + \beta\gamma n$ grows quadratically so that the threshold t is reached near the deterministic crossing time $m(t)$ in which $\xi_{m(t)} \approx t$, i.e.

$$m(t) = \frac{-\beta\gamma + \sqrt{\beta^2\gamma^2 + 2\beta t}}{\beta} = \sqrt{\frac{2t}{\beta}} - \gamma + o(1).$$

By the functional central limit theorem, the deviation of $\nu(t)$ about $m(t)$ is approximately normal. Linearization of $Z_{\nu(t)} \approx t$ around $m(t)$ yields

$$\nu(t) \mathrel{\dot\sim} \mathrm{Normal}\left(m(t), \; \frac{\sigma^2 \, m(t)}{\big(\alpha + \beta \, m(t)\big)^2}\right) \; = \; \mathrm{Normal}\left(\sqrt{\frac{2t}{\beta}} - \gamma, \; \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right),$$

where we have used the fact that $\alpha + \beta m(t) \sim \beta m(t) \sim \sqrt{2\beta t}$ for $t \gg 0$. Thus, in regime 2 the mean hitting time scales like $\sqrt{t}$ (rather than linearly in t) and the variance shrinks at rate $t^{-1/2}$ (rather than growing at t).

A.2 Additional Figures

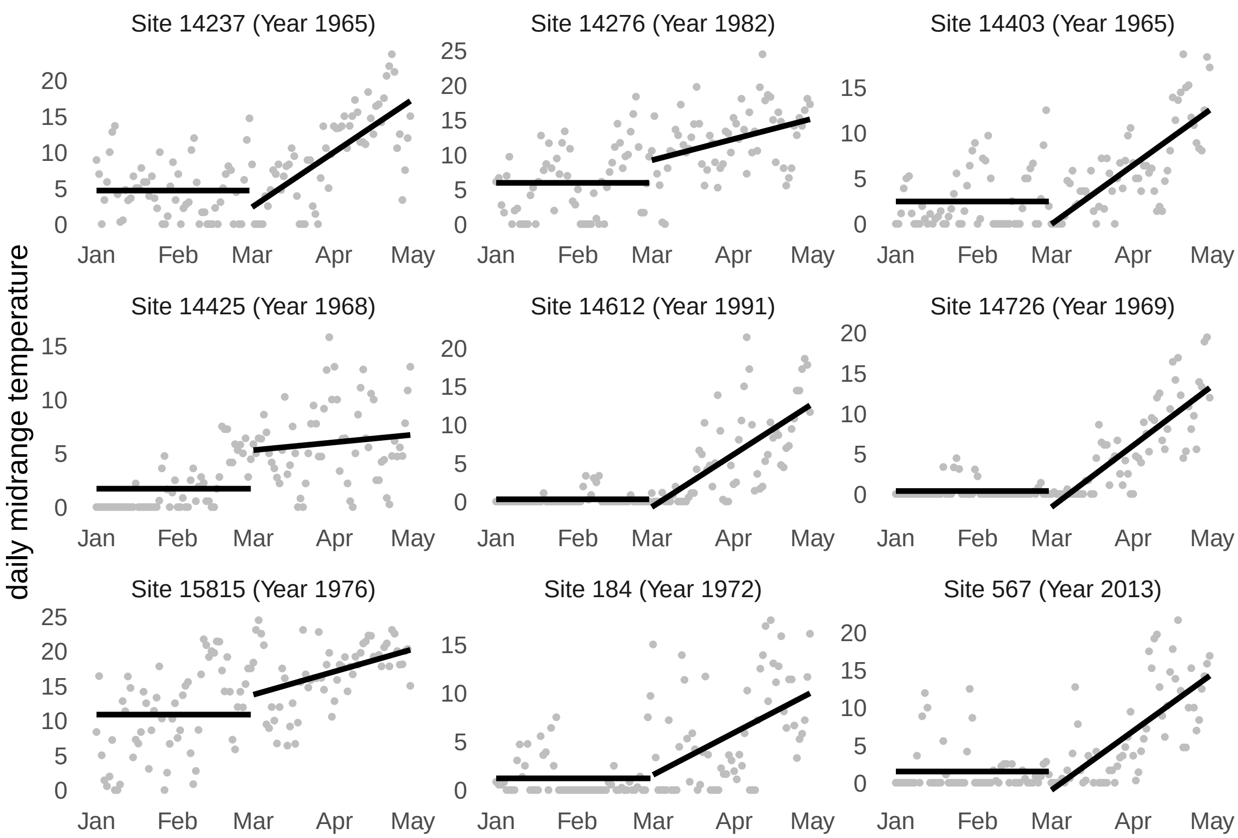

Figure 4: Springtime daily midrange temperatures (points) with α (mean from Jan to Feb) and β (slope from Mar to Apr) for nine randomly selected sites from lilac-honeysuckle dataset (Rosemartin et al. 2015) as measured by GHCND stations nearby. Temperatures below 0 are set to 0.

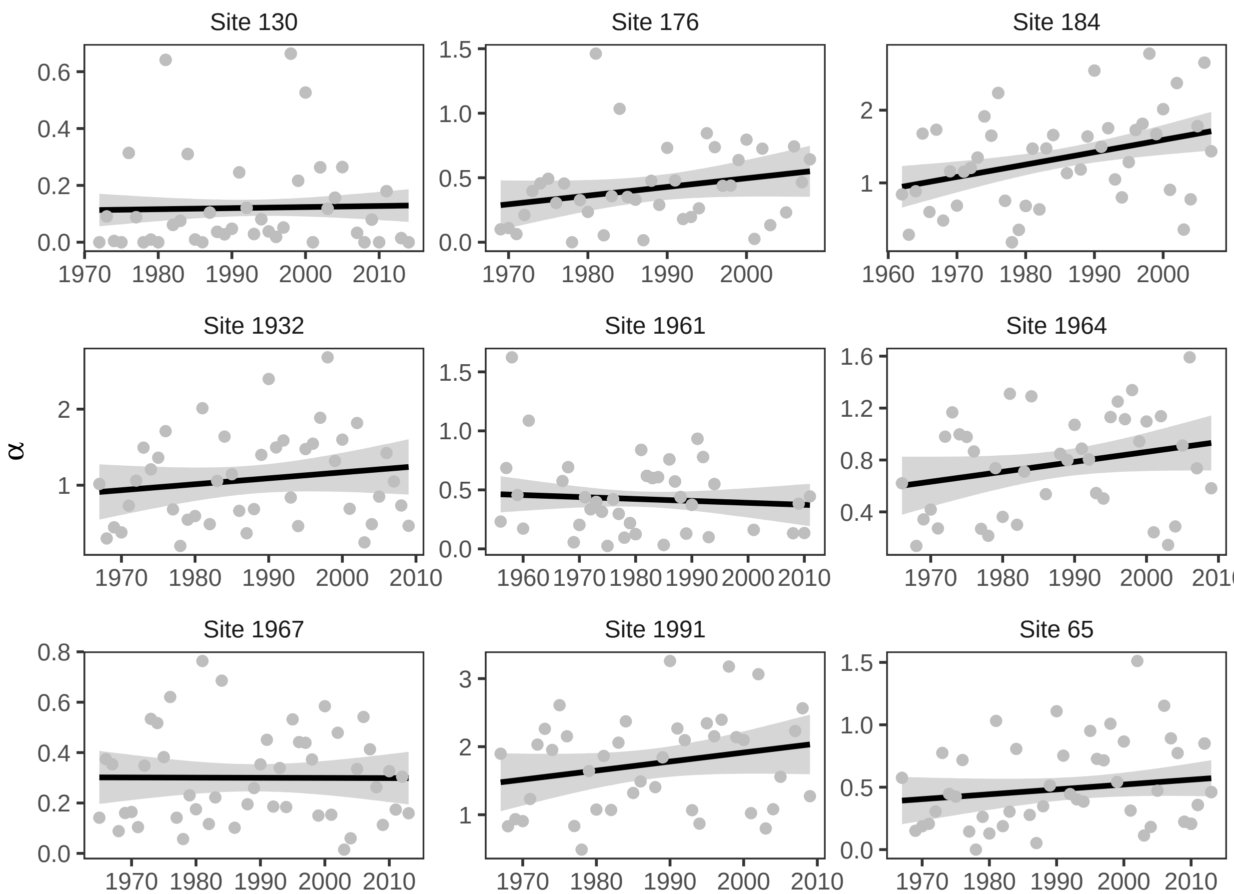

Figure 5: Changes in α for the nine sites from lilac-honeysuckle data (Rosemartin et al. 2015) with longest running records as as measured by GHCND stations nearby.

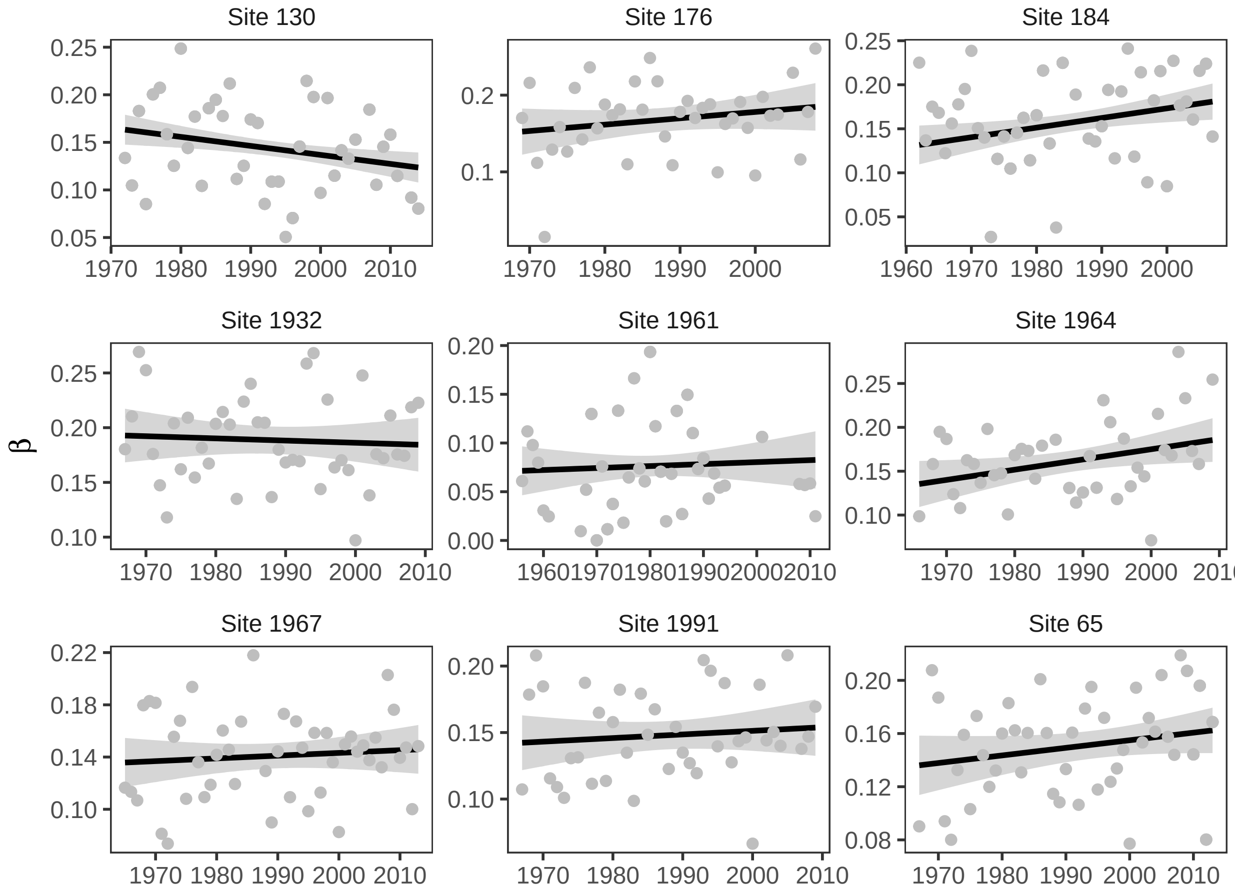

Figure 6: Changes in β for the nine sites from lilac-honeysuckle data (Rosemartin et al. 2015) with longest running records as as measured by GHCND stations nearby.

A.3 Simulations

We run two simulations to study the results from Sections 2 and 3. The first checks the asymptotic approximation from Section 2. The second checks the interpretation of the data from Section 3.

Simulation 1

The first simulation verifies the accuracy of the two asymptotic normal approximations for the hitting time

$$\nu(t) = \min\left\{n \geq 1 : \sum_{i=1}^{n} X_i > t\right\}$$

under the model

$$X_i = \alpha + \beta i + \varepsilon_i, \qquad \varepsilon_i \sim \text{Normal}(0, \sigma^2).$$

For simulation, we set $\sigma = 20$ and generate daily temperatures until the threshold t is reached, recording the corresponding hitting time $\nu(t)$. We repeat this procedure $R = 10{,}000$ times for thresholds $t = \{1000, 2000\}$, $\alpha = \{2, 4\}$ and $\beta = \{0, 0.1\}$. When $\beta = 0$ the simulation matches the winter regimes, and when $\beta = 0.1$, it matches the spring regime. The simulated distribution of $\nu(t)$ is represented in Figure 6 below by the histograms.

We then compare these simulations with the theoretical approximations, we standardized each simulated hitting time using the relevant asymptotic mean and standard deviation. When $\beta = 0$, we use the winter regime approximation

$$\nu(t) \overset{\cdot}{\sim} \text{Normal}\left(\frac{t}{\alpha}, \frac{\sigma^2 t}{\alpha^3}\right).$$

When $\beta > 0$, we use the spring regime approximation

$$\nu(t) \overset{\cdot}{\sim} \text{Normal}\left(\sqrt{\frac{2t}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2}, \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right).$$

For each setting we computed the standardized variable

$$z = \frac{\nu(t) - \mu}{\text{sd}},$$

plotted its histogram, and overlaid the standard normal density. Across both regimes, the standardized histograms align closely with the $\text{Normal}(0, 1)$ curve, and the agreement improves as t increases, providing a visual confirmation of the derived asymptotic distributions.

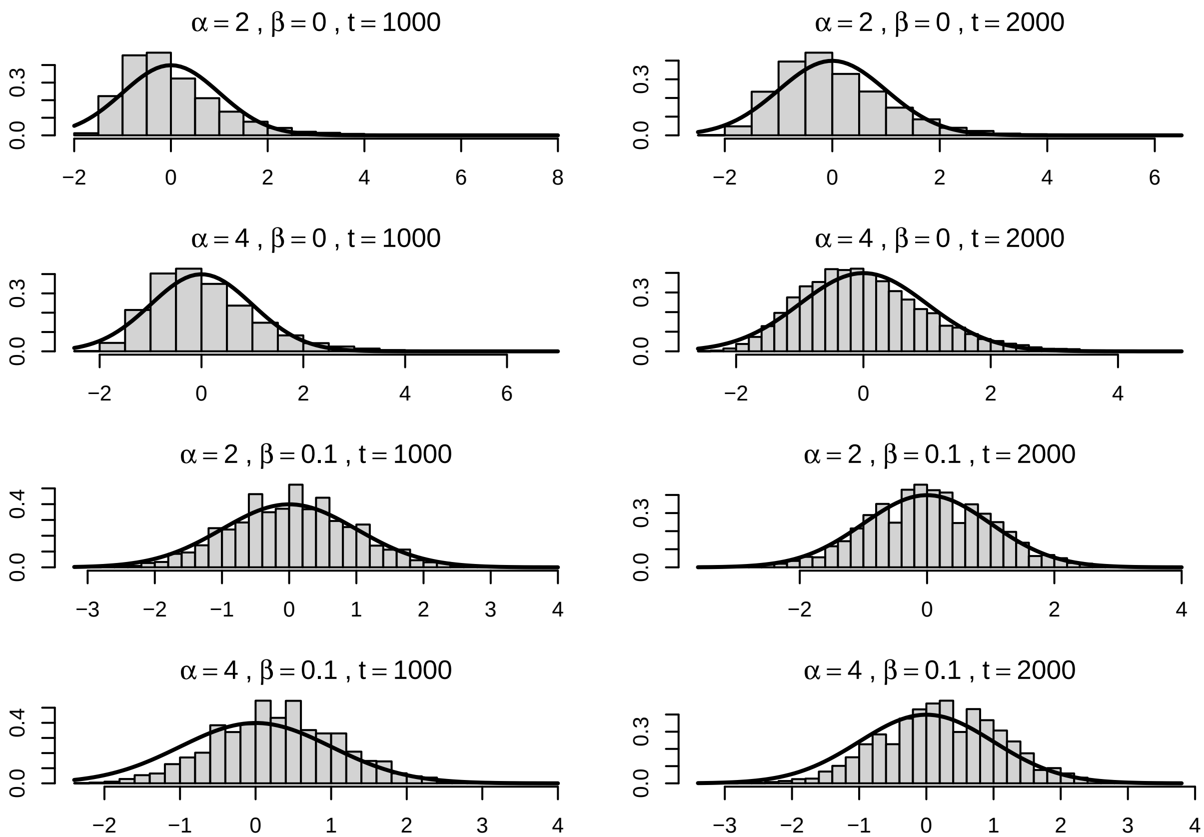

Figure 7: Simulations comparing the distribution of bloom dates under two regimes: stationary (top) and non-stationary (bottom).

Simulation 2

To verify our interpretation of the data in Section 3, we simulate the model from section. In each replicate we generated daily temperatures

$$X_i = \mu_i + \varepsilon_i, \qquad \varepsilon_i \sim \text{Normal}(0, \sigma^2),$$

with $\sigma = 20$, where the deterministic component follows

$$\mu_i = \begin{cases} \alpha, & i = 1, \ldots, 90 \quad \text{(January–February)}, \\ \alpha + \beta(i - 90), & i = 91, \ldots, 180 \quad \text{(March–April)}. \end{cases}$$

Bloom occurs when cumulative temperature first exceeds a threshold t, i.e.

$$\nu(t) = \min\Big\{n \geq 1 : \sum_{i=1}^{n} X_i > t\Big\}.$$

We simulated $R = 10{,}000$ independent realizations of $\nu(t)$ for each combination of $\alpha \in \{4, 8, 10\}$ and $\beta \in \{0.2, 0.4, 0.8\}$, and for thresholds $t \in \{1000, 2000\}$. For each parameter pair we report the Monte Carlo mean and standard deviation of $\nu(t)$ (Tables 4a–4d), arranged with rows indexing α and columns indexing β.

The results reproduce the qualitative patterns observed in the lilac application. Mean bloom dates decrease as either α increases or β increases (Tables~4a and 4c). Within a fixed α row, increasing β reduces the standard deviation of bloom dates (Tables~4b and 4d), consistent with the prediction that that increasing average temperatures diminish the influence of noise on the threshold-crossing time. Holding β fixed, the standard deviation is often larger at higher α (particularly for $\beta \in \{0.4, 0.8\}$), consistent with the regime-shift intuition that earlier crossings can occur in a flatter portion of the seasonal cycle where microclimate variability has greater influence.

Table 4: Simulated mean and standard deviation of the bloom date $\nu(t)$ across $R = 10{,}000$ Monte Carlo replicates, for thresholds $t \in \{1000, 2000\}$. Rows index the winter level α and columns index the spring warming rate β.

(a) mean, $t = 1000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	151.66	136.83	124.86
8	114.87	111.27	106.85
10	98.45	97.20	95.72

(c) mean, $t = 2000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	199.54	171.15	149.16
8	169.81	152.43	137.37
10	156.22	143.40	131.34

(b) sd, $t = 1000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	15.48	10.42	7.21
8	16.53	13.27	10.50
10	15.94	14.45	12.71

(d) sd, $t = 2000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	10.96	7.22	4.84
8	10.93	7.50	5.11
10	10.69	7.66	5.36